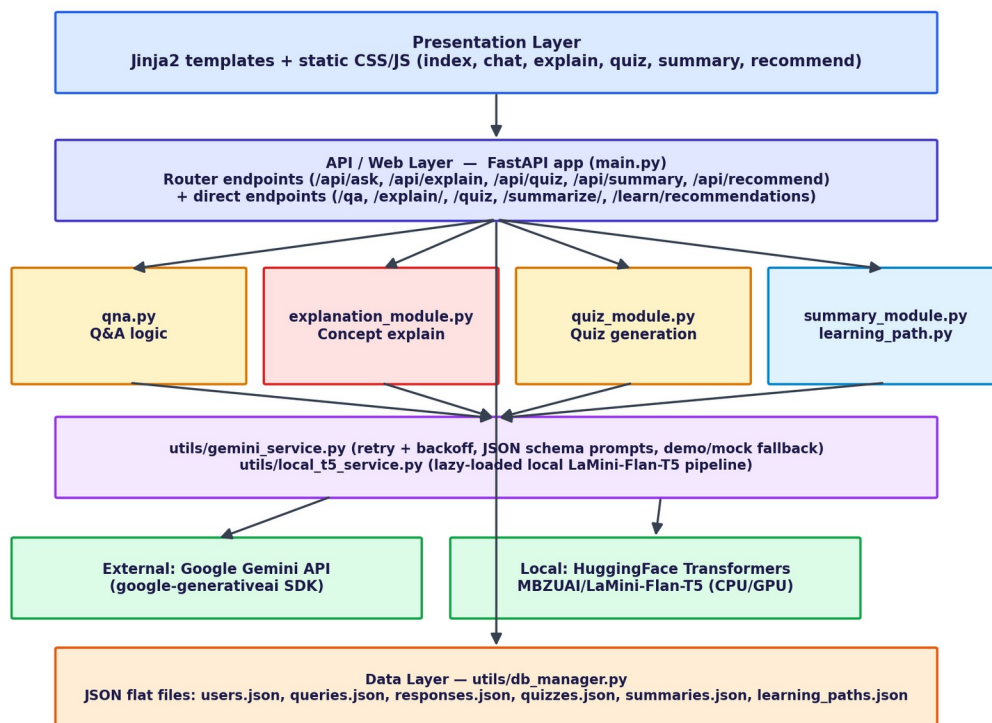


10. Solution Architecture

Date	28 June 2026
Team ID	SWTID-2026-8135
Project Name	EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

High-level view of EduGenie's Presentation, API, Service and Data layers, and how they connect to Google Gemini and the local LaMini-Flan-T5 model.

EduGenie - Solution Architecture



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the home, chat, explain, quiz, summary, recommend, about and contact pages; handles client-side interactivity (timers, copy buttons, theme toggle, markdown download).	Jinja2 templates, HTML5, CSS (glassmorphism), vanilla JavaScript
API / Web Layer	FastAPI app (main.py) exposing router-based endpoints (/api/ask, /api/explain, /api/quiz, /api/summary, /api/recommend) and direct RESTful endpoints (/qa, /explain/, /quiz, /summarize/, /learn/recommendations); mounts static files and	FastAPI, Uvicorn, CORS middleware

Component Name	Description / Role in Architecture	Technologies Used
	serves HTML pages.	
Service / Core Logic Layer	Feature-specific modules (qna.py, explanation_module.py, quiz_module.py, summary_module.py, learning_path.py) that build prompts, call Gemini or the local T5 pipeline, parse/validate JSON responses, and fall back to mock data on failure.	Python, Pydantic schemas (models/schemas.py)
AI Integration Layer	utils/gemini_service.py resolves the best available Gemini model at startup and wraps calls with retry + exponential backoff; utils/local_t5_service.py lazily loads and runs the local LaMini-Flan-T5 pipeline in a background thread.	google-generativeai SDK, HuggingFace transformers, PyTorch
Data Layer	utils/db_manager.py reads/writes JSON collections for users, queries, responses, quizzes, summaries and learning paths, and seeds a default demo user at startup.	JSON flat files (data/*.json)
Deployment	Containerized with a slim Python base image; runs the Uvicorn ASGI server on port 8000.	Docker (python:3.10-slim)